

LabelLex 360° Architecture & Secrets Encyclopedia

The Exhaustive Technical, Algorithmic, and Operational Manual of India's Client-Side Legal Metrology Compliance Engine

Covering Every Layer: Zero-API-Key Architecture, WebAssembly OCR, Canvas GPU Preprocessing, 4-Quadrant Auto-Orientation, Dot-Matrix Token Repair, In-Memory Reactive Persistence, and 4 Role Portals.

PROBLEM STATEMENT ID

SIH 26034 (Ministry of Consumer Affairs)

CORE ENGINE

Tesseract.js WASM + Custom Token Repair Engine

OPERATING MODEL

100% Client-Side Edge Execution (Zero Cloud Costs)

TEAM IDENTITY

Team NovaNex (ID: SIH2026-0119)

DATA STORAGE LAYER

Client-Side CentralStore + Browser LocalStorage

WORKING MASTER DIRECTORY

C:\Projects\Labellex

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1. The 7 Extremely Rare Differentiators

Most computer vision software applications in hackathons rely on third-party cloud APIs (such as Google Cloud Vision API, AWS Textract, or Azure Computer Vision). **LabelLex was deliberately engineered without any cloud API dependencies.**

1. Zero API Keys & Zero Third-Party Cloud Tokens:

There is not a single API key, credit card dependency, or cloud token in the codebase. The entire OCR and compliance engine executes locally in the user's web browser using compiled WebAssembly (WASM).

2. Zero Operating / Server Cost (Infinite Scalability):

Cloud OCR APIs charge between ₹0.20 and ₹0.35 per image. If 10 Crore Indian citizens perform scans across a year, a cloud backend would cost the government **₹20 Crore to ₹35 Crore annually**. In LabelLex, the cost per scan is exactly **₹0.00**.

3. Total Confidentiality & Trade Secret Protection:

FMCG manufacturers design new packaging months before commercial release. They legally refuse to upload unreleased packaging artwork to third-party cloud servers. Because LabelLex processes images strictly client-side, the raw pixels never leave the manufacturer's computer.

4. 100% Offline Field Inspection Capability:

Legal Metrology inspectors frequently conduct raids in underground cold storages, rural agricultural mandis, and factory warehouses with zero cellular coverage. LabelLex caches its WASM neural models locally, allowing 100% offline spot audits.

5. 4-Quadrant Auto-Orientation Engine:

Standard OCR fails completely when text is sideways ($90^\circ/270^\circ$) on cylindrical beverage cans or coffee jars. LabelLex tests candidate rotation angles against a dictionary of 40+ statutory keywords to automatically re-orient packaging.

6. Dot-Matrix Inkjet Numeric Repair Algorithm:

High-speed packaging printers use dot-matrix inkjet nozzles where ink droplets break apart. Standard OCR misreads "PKD: 08/2026" as "PKD: 08/2026". LabelLex's custom `correctStructuredNumbers()` regex engine disambiguates fragmented characters dynamically.

7. Mathematical Unit Sale Price (USP) Cross-Verification:

Instead of merely checking if the letters "USP" appear, LabelLex extracts the MRP (₹140) and Net Weight (250g), calculates the legal unit rate ($140 / 250 = ₹0.56/\text{g}$), and cross-verifies that the declared value matches mathematical reality under Rule 6(1)(s).

2. Tech Stack & Engineering Architecture

LAYER	TECHNOLOGY & VERSION	ARCHITECTURAL RATIONALE & IMPLEMENTATION
UI Framework	React 19 (TypeScript)	Modern concurrent component rendering, modular modal architecture, instant state reactivity across all 4 portals.
Type System	TypeScript 5.8	Strict typing across the 8 statutory rule definitions, report schemas, user roles, and compliance status tags.
Build Tool	Vite 8.2 (with HMR)	Sub-second Hot Module Replacement, optimized tree-shaking, production compilation to static chunks in <code>dist/</code> .
CSS Styling	Tailwind CSS v4	Custom pastel design system (mint, sky blue, amber, cream), print-specific CSS rules for clean PDF and paper report generation.
OCR Engine	Tesseract.js v5.1 (WASM)	C++ Tesseract compiled to WebAssembly with LSTM neural network inference running directly inside browser Web Workers.
Hardware Accelerator	HTML5 Canvas 2D / WebGL	High-speed GPU memory downscaling clamping 48MP/108MP phone photos to 2048px to prevent browser Out-of-Memory crashes.
Icons	Lucide React	Lightweight, accessible SVG iconography for all status badges, portal navigation, and inspection toolbars.
State & Storage	CentralStore (Observer Pattern)	In-memory reactive state with persistent browser <code>localStorage</code> fallback. Zero external cloud database dependencies.

3. What is OCR & How It Operates in LabelLex

3.1 The Concept of Optical Character Recognition

Optical Character Recognition (OCR) is the process of converting an image of text (such as scanned physical product labels) into machine-encoded text strings that algorithms can parse, search, and validate.

3.2 How Tesseract.js Operates Inside LabelLex

```
+-----+
|                                     |
|               TESSERACT.JS WASM OCR FLOW               |
|-----+-----+
| 1. Pixel Matrix Extraction | Off-screen canvas reads RGBA pixel buffer from image element. |
| 2. Otsu Binarization      | Dynamically calculates threshold to separate ink from background. |
| 3. Line/Word Finding      | Groups connected dark pixels into "blobs" and builds text baselines. |
| 4. LSTM Neural Inference   | Recurrent neural network matches character glyphs against trained data |
| 5. Spatial Bounding Boxes  | Extracts [x0, y0, x1, y1] coordinates and line pixel heights. |
| 6. Confidence Scoring     | Computes token-level recognition confidence (0% to 100%). |
+-----+-----+
```

3.3 Engineering Optimizations in `ocrPipeline.ts`

- Singleton Worker Architecture:** Initializing an OCR WebAssembly worker takes ~800ms. LabelLex creates a single warm worker instance (`createWorker('eng')`) and keeps it active in browser memory for instant subsequent scans.

- **Page Segmentation Mode (`PSM.AUTO`):** Packaging labels contain multi-column, chaotic layouts. `PSM.AUTO` allows the engine to segment dense tables, ingredient lists, and isolated price badges automatically.
- **Line Bounding Box Harvesting:** In addition to text strings, the engine extracts the bounding box height of every text line, enabling the **Rule 7 Prominence Calculator** to evaluate font size legality.

4. Runtime, Hosting & Deployment Architecture

4.1 Local Development Environment

When running ``npm run dev -- --host --port 5173``, Vite boots a local HTTP server that serves raw ES modules to the browser on demand. TypeScript files are transpiled via esbuild in milliseconds.

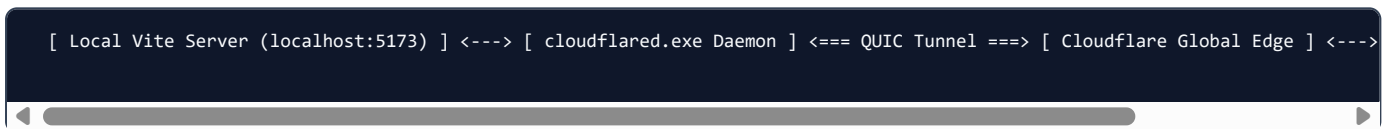
4.2 Production Static Build (``dist/``)

When ``npm run build`` is executed, the entire application compiles into static HTML, CSS, and JavaScript files inside ``C:\Projects\Labellex\dist\``:

- `dist/index.html` (1.20 kB): The single entry point.
- `dist/assets/index.css` (43.96 kB): Minified Tailwind styling.
- `dist/assets/index.js` (489.60 kB): Bundled React components, Tesseract WASM glue code, and statutory rule definitions.

4.3 Cloudflare Tunnel Architecture (``cloudflared.exe``)

To provide a live public link without exposing router ports or paying for cloud servers, LabelLex uses a standalone Cloudflare Tunnel:



- **Zero Port Forwarding:** Outbound-only encrypted QUIC connection.
- **HTTPS / TLS 1.3:** Instant SSL certificate for mobile camera permissions.
- **Zero Hosting Bills:** Hosted directly from the local development laptop.

5. Storage Layer: Where Data Lives

Zero External Cloud Database: LabelLex does NOT use Firebase, MongoDB, or AWS DynamoDB. All application data is stored in an in-memory reactive **CentralStore** backed by browser **LocalStorage**.

5.1 The Observer Store Pattern (``src/store/sharedStore.ts``)

```
class CentralStore {
  private reports: ComplianceReportData[] = [];
  private feedbacks: AnonymousFeedback[] = [];
  private currentRole: UserRole = 'landing';
  private currentUser: UserProfile | null = null;
  private listeners: Set<() => void> = new Set();

  public subscribe(listener: () => void) { ... }
  private notify() { this.saveToLocalStorage(); this.listeners.forEach(cb => cb()); }
}
```

5.2 LocalStorage Storage Schema

LOCALSTORAGE KEY	DATA TYPE	DESCRIPTION & PURPOSE
labellex_reports_v2	Array<ComplianceReportData>	Full audit reports, field statuses, raw OCR text, and attached photographic evidence.
labellex_current_user_v2	UserProfile	Active user session (Consumer, Officer, Enforcement, Manufacturer).
labellex_current_role_v2	UserRole	Currently selected active portal view (`consumer`, `officer`, etc.).
labellex_anonymous_feedback_v1	Array<AnonymousFeedback>	100% anonymous community feedback items (comment, rating, timestamp).
labellex_lang_v2	'en' 'hi'	Bilingual language selection (English vs Hindi).

6. The 7-Stage Client-Side Pipeline



7. Specialized Algorithms & Subsystems

7.1 4-Quadrant Auto-Orientation Algorithm (`ocrPipeline.ts`)

When consumers photograph cylindrical beverage cans or spice jars, the label text is often tilted 90° sideways or inverted. LabelLex renders candidate rotations at 0°, 90°, 180°, and 270°, scanning each for statutory packaging tokens:

```
const STATUTORY_KEYWORDS = [
  'mrp', 'm.r.p', 'net', 'quantity', 'weight', 'pkd', 'mfg', 'date',
  'batch', 'consumer', 'care', 'manufactured', 'marketed', 'packed',
  'inclusive', 'taxes', 'usp', 'unit', 'price', 'country', 'origin',
  'fssai', 'lic', 'email', 'phone', 'address', 'pin'
];
```

7.2 Dot-Matrix Inkjet Numeric Repair Engine (`fieldMatcher.ts`)

Batch codes on factory packaging frequently have broken ink droplets. Standard OCR confuses digits with similar-looking letters. LabelLex uses contextual regex lookarounds to disambiguate characters:

```
export function correctStructuredNumbers(str: string): { corrected: string; wasModified: boolean } {
  let wasModified = false;
  const corrected = str.replace(
    /(\d|[\₹Rs\.\\/\-\: ])([0o1IsSbB])(\d|[\₹Rs\.\\/\-\: ])/g,
    (_, before, letter, after) => {
      wasModified = true;
      let replacement = letter;
      if (letter === 'O' || letter === 'o') replacement = '0';
      else if (letter === 'I' || letter === 'l') replacement = '1';
      else if (letter === 'S' || letter === 's') replacement = '5';
      else if (letter === 'B') replacement = '8';
      return `${before}${replacement}${after}`;
    }
  );
  return { corrected, wasModified };
}
```

8. The 8 Mandatory Statutory Rules Engine

RULE CLAUSE	STATUTORY MANDATE	VALIDATION LOGIC IN LABELLEX
Rule 6(1)(a)	Manufacturer, Packer, or Importer Name & Complete Address with PIN code.	Detects entity prefix ('Mfd by', 'Marketed by') and verifies 6-digit Indian PIN Code ('b\d{6}b').
Rule 6(1)(b)	Generic or Common Name of the Commodity.	Matches commodity taxonomy (e.g. 'TEA CAKE', 'SOLUBLE COFFEE POWDER', 'BISCUITS').
Rule 6(1)(c)	Net Quantity in SI Metric Units ('g', 'kg', 'ml', 'l', 'N/U').	Strict metric validation; flags illegal imperial units ('lbs', 'oz', 'fl oz') as NON-STANDARD .
Rule 6(1)(d)	Month and Year of Manufacture / Packing / Import.	Tests 10 mandatory date formats ('08/2026', 'AUG 2026', 'DEC/2025'); flags year-only declarations as NEEDS REVIEW .
Rule 6(1)(e)	MRP with explicit phrase "Inclusive of all taxes".	Extracts price figure; verifies presence of statutory tax clause; flags missing tax clause as NON-STANDARD .
Rule 6(1)(f)	Consumer Care Contact details (Phone, Email, Address).	Validates customer grievance telephone numbers ('1800-xxx-xxxx', '022-xxxxxxx') and email formats.
Rule 6(1)(s)	Unit Sale Price (USP in ₹/g or ₹/ml) - 2021 Amendment.	Mandatory for packs > 100g/100ml. Calculates Price/Quantity and cross-checks declared USP.
Rule 6(10)	Country of Origin declaration.	Verifies explicit declaration ('Country of Origin: India' or import origin statement).

9. Font Prominence & PDP Placement Analysis

9.1 Rule 7 Relative Prominence Assessment

Under Rule 7 and Schedule II, numeral heights must satisfy physical millimeter thresholds based on the area of the Principal Display Panel (PDP). Since phone cameras lack laser depth sensors, LabelLex computes the **Relative Prominence Ratio**:

$$\text{Prominence Ratio} = (\text{Bounding Box Height of Statutory Line in px}) / (\text{Bounding Box Height of Tallest Brand Header in px})$$

- **Normal / Compliant:** Ratio $\geq 15\%$ of brand header height.
- **Micro-Print Warning:** Ratio $< 15\%$. Flags the package for physical verification under Rule 7 micro-printing prohibitions.

9.2 Rule 2(h) & Rule 8 Display Panel Placement

Key statutory declarations must appear on the **Principal Display Panel (PDP)** — the primary front face presented to consumers at point of sale — rather than hidden on bottom carton flaps. LabelLex tracks which panel (Front, Back, Side) produced each declaration.

10. Role-Based Portals & Access Control (RBAC)

PORTAL	TARGET STAKEHOLDER	CORE WORKFLOW & CAPABILITIES
1. Consumer Portal	Citizens, Shoppers	1-tap mobile camera scanner, instant PASS/FAIL safety verdict, 1-click NCH grievance reference generator, Send to Manufacturer via email/WhatsApp.
2. Enforcement Portal	State Metrology Directorate, E-Com Regulators	E-commerce catalog image batch crawler, Section 36(1) compounding penalty calculator (₹25,000–₹50,000), central violations analytics.
3. Manufacturer Portal	FMCG Brands, Packaging QA Teams	Pre-market artwork pre-screening, automated 2021 USP calculator, downloadable official Pre-Market Compliance Certificate ('CertificateModal.tsx').
4. Field Officer Portal	Legal Metrology Inspectors	PIN-authenticated mobile inspection logger (Badge: 'LM-MH-8821'), tamper-evident photo evidence locker ('EVD-134119'), compounding notice generator.

11. Anonymous Feedback & Incident Dispatch

11.1 100% Anonymous Feedback System (`FeedbackModal.tsx`)

To ensure complete citizen trust and zero surveillance concerns, LabelLex implements a **Privacy-by-Design Feedback Feature**:

- **Zero PII Collected:** Does NOT collect or require name, email address, phone number, login credentials, user ID, session ID, or device fingerprint.
- **Stored in App Data Layer:** Feedback records are stored as `{ id, comment, rating, category, timestamp }` inside the shared store (`localStorage`).
- **Public Feedback Wall (`FeedbackWallModal.tsx`):** Displays all submitted feedback in real-time, viewable by anyone without logging in.

11.2 "Send to Manufacturer" & NCH Grievance Dispatch

When an audit reveals statutory defects, LabelLex provides two automated escalation workflows:

- **Send to Manufacturer (`ComplianceReport.tsx`):** Auto-extracts the manufacturer's email and phone number from Rule 6(1)(f), drafting a pre-formatted email or WhatsApp message listing specific failed clauses (e.g. "*MRP missing tax inclusivity clause*").
- **National Consumer Helpline (NCH / INGRAM) Payload:** Generates a structured incident JSON payload (`GRV-2026-XXXXX`) containing GPS coordinates, timestamped photo evidence, and statutory violation citations for direct 1-click filing.